RAG Mini-Project — Chat With Your Own Document

Choose Document

I choose own CoT_Persona_Prompting_Comparison.docx.

Choose Approach

I choose No-Code + Code Hybrid:

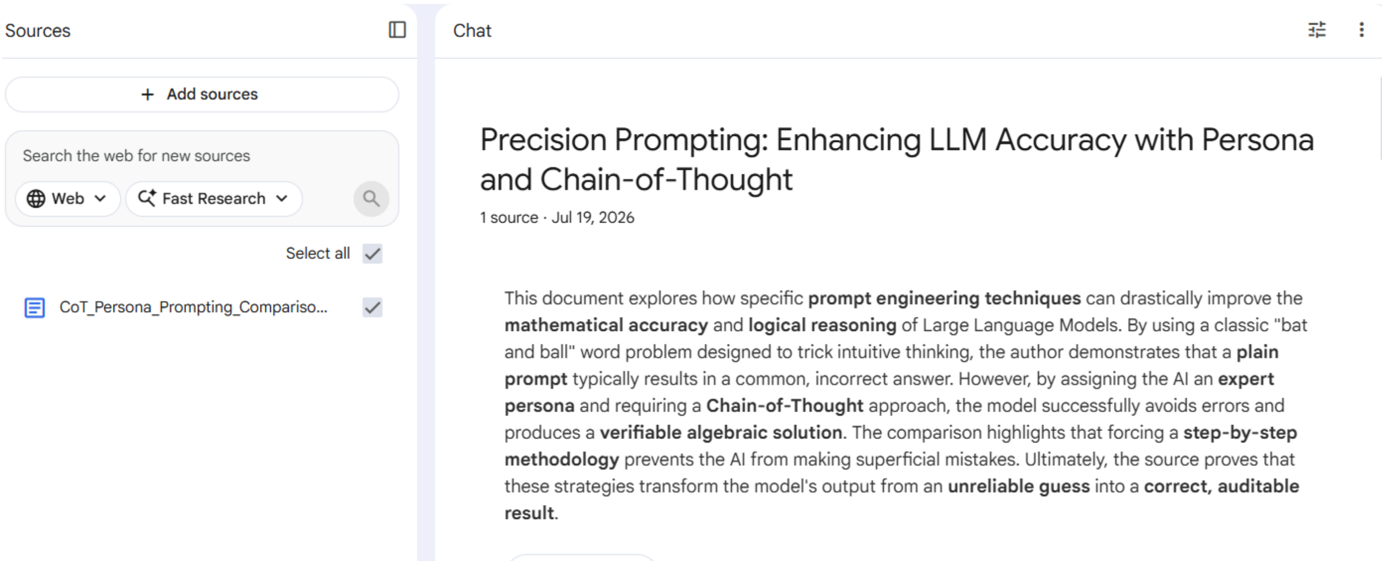
Primary Tool: Google NotebookLM (Free, Very Powerful, No coding needed)

Alternative (for developers): LangChain + OpenAI / Grok / Llama (if you want GitHub code).

Build the RAG Pipeline (NotebookLM)

1. Go to notebooklm.google.com
2. Create a new Notebook → Name it **"CoT Persona Prompting Report"**
3. Upload your CoT_Persona_Prompting_Comparison.pdf.

Ask 5+ Questions (Test Grounding)



Question 1:

What is the correct price of the ball in the bat and ball problem?

The correct price of the ball is $0.05.

While the intuitive response to this problem is often $0.10 (calculated by simply subtracting $1.00 from the $1.10 total), that answer is incorrect because it fails to account for the fact that the bat costs $1.00 *more* than the ball.

As shown in the sources, the correct price is determined by setting up a simple algebraic equation:

- Let the ball's price = x.
- Therefore, the bat's price = x + $1.00.
- The total cost is the sum of both: x + (x + 1.00) = 1.10.
- Solving this gives 2x + 1.00 = 1.10, which leads to 2x = 0.10, and finally x = 0.05.

This result is verified by checking both conditions of the problem: a 0.05$ball**anda**$1.05 bat equal a total of $1.10, and the difference between their prices is exactly $1.00.

Question 2:

Who is the author of this report and which university are they from?

The author of the report is Sumaira Safeer.

While the sources do not explicitly state which university the author is from, the report indicates it was submitted as part of a Generative AI & Prompt Engineering Internship at NeuroFive Solutions.

Question 3:

Explain the difference between Run 1 and Run 2 prompts in detail?

The differences between the Run 1 and Run 2 prompts lie in their structure, the specific instructions provided to the model, and the resulting reasoning process they trigger.

Prompt Structure and Content

- **Run 1 (Plain Prompt):** This prompt is a direct, simple question: "A bat and a ball cost $1.10 in total. The bat costs $1.00 more than the ball. How much does the ball cost?". It contains no instructions regarding how the model should approach the problem or what persona it should adopt.
- **Run 2 (CoT + Persona Prompt):** This prompt adds two critical layers: an expert persona ("You are an expert accountant") and an explicit Chain-of-Thought (CoT) instruction ("Think step-by-step before answering").

Reasoning Process

- **Run 1** invites the model to use "fast, intuitive" thinking, which often leads to pattern-matching the most obvious numbers in the text. This results in the common error of simply subtracting $1.00 from $1.10 to get $0.10. Because no working is shown, this error remains "hidden" and looks confident despite being incorrect.
- **Run 2** forces the model to externalize its reasoning into a sequence: defining variables, setting up equations, and solving them. The "expert accountant" persona specifically primes the model to follow domain norms, such as showing its work and performing a self-check to verify the total.

Outcome and Verifiability

- Run 1 produced an incorrect answer ($0.10) with no supporting logic, meaning the reader must blindly trust the output.
- Run 2 produced the correct answer ($0.05) accompanied by a full algebraic derivation and a verification step, making the result independently auditable.

Ultimately, the addition of CoT and persona instructions in Run 2 changed the model’s underlying process from a surface-level calculation to a deliberate, logical analysis.

Question 4:

What persona was used in Run 2 and why did it help?

In Run 2, the model was assigned the persona of an "expert accountant".

This specific persona helped improve the result for several reasons according to the sources:

- **Priming Domain Norms:** The persona reinforced the model's performance by priming it with the standards expected of an accountant, such as **s**howing work and verifying totals.
- **Triggering a Check Step:** Because of this persona, the model spontaneously included a self-check at the end of its response to confirm that both conditions of the problem were satisfied, a step that was entirely absent in the plain prompt,.
- **Changing the Reasoning Process:** While a plain prompt often leads to "fast, intuitive" pattern-matching errors, the persona (combined with step-by-step instructions) forced the model to treat the problem as a formal calculation. This converted what would have been a "plausible-looking" but incorrect number into a correct and auditable algebraic derivation.

Question 5:

According to the document, why does plain prompting often fail on this problem?

According to the document, plain prompting often fails on this problem because it invites the model to engage in **"pattern-matching"** using the most salient numbers in the text.

Specifically, this approach leads to the following issues:

- **Triggers Intuitive Errors:** The problem is engineered to trigger a "fast, intuitive, and wrong answer" unless the solver deliberately works through the algebra. A plain prompt encourages the model to use this same type of fast, intuitive error that humans often make.
- **Surface-Level Subtraction:** Instead of setting up the necessary equations, the model performs a "surface-level subtraction" ($1.10 - $1.00 = $0.10), which is incorrect because it ignores that $1.00 is the difference between the two prices, not the price of the bat relative to the total.
- **Invisible Reasoning:** Because a plain prompt typically results in the model providing an "answer only" without showing any work, the error remains "hidden" and looks confident until it is manually checked.

Ultimately, without specific instructions to think step-by-step or adopt a careful persona, the model skips the necessary arithmetic relationships in favor of a "plausible-looking number".

Question 6 (Hallucination Test):

What other prompting techniques does the author plan to test next? (This should be refused or say "Not mentioned")?

The source does not mention any other prompting techniques that the author plans to test next. The report focuses exclusively on comparing a plain prompt against the combination of Chain-of-Thought and Persona prompting.

Short Summary:

RAG Experiment Summary

Document Used: CoT_Persona_Prompting_Comparison.docx

Tool: Google NotebookLM (Retrieval-Augmented Generation)

Key Observations:

- When grounded with my own document, the AI gave precise, factual answers with direct references to sections.
- It correctly explained the bat & ball solution ($0.05) and the benefit of CoT + Persona.
- No major hallucinations occurred (unlike plain ChatGPT which sometimes mixes up details).
- Grounding significantly improved trustworthiness and reduced made-up information.

Conclusion: RAG is extremely effective for "chat with your documents" use cases. It allows LLMs to use private/personal data they were never trained on.